r/Futurology

Future of climate change: COP29 & Tech companies investing in nuclear energy to power AI

Posted by u/OkToe7809 • • 0 points • 17 comments

Hey guys, 2 key developments this year in the fight against climate change:

* Big Tech (Google, Amazon) invests in nuclear energy (small modular reactors SMRs) to power AI, to come online in 2030s
* Flagging engagement in COP29 by government leaders

NGOs have historically struggled to solve issues (United Nations, WTO, WHO), needing too much consensus, plus it's hard for governments to make a business case to enact policies to fight climate change (what it comes down to at the end of the day). Whereas private sector will solve problems when there's an economic incentive, as the pursuit of AI technology has now created for Big Tech to scale energy supply. Curious how to create a business case to clean up the environment, ie, with AI robots (I hated typing that sentence, but in reality that's how these things actually get done).

What would it look like if this trend continues? Or what do you think the most plausible scenarios are for humanity to actually survive climate change?

I asked ChatGPT, curious your thoughts:

Here's a timeline for **Big Tech-led developments in nuclear energy and AI robots** to tackle climate change and optimize global manufacturing, contrasting the slower progress of intergovernmental efforts like COP summits. This timeline is speculative but based on current technological and corporate trajectories:

2024-2025: Foundation & Investment Phase

* **Big Tech Invests in SMRs (Small Modular Reactors)**:
   * Companies like Google and Microsoft continue investing in nuclear startups developing SMRs, which promise safer, scalable, and more cost-effective nuclear energy solutions.
   * Early partnerships form between tech giants, energy firms, and advanced material startups to fast-track SMR construction.
   * Pilot projects for AI-managed energy distribution systems begin, optimizing energy grids for reduced waste and better integration of renewables with nuclear.
* **AI Robots for Green Manufacturing**:
   * AI-powered robots begin rolling out in manufacturing for tasks like precision recycling, reducing material waste, and creating energy-efficient supply chains.
   * Companies like Tesla, Amazon, and Apple scale up R&D in autonomous robotics for climate initiatives, focusing on solar panel deployment, reforestation, and carbon capture.

2026-2028: Early Deployment & Optimization

* **SMR Construction Advances**:
   * The first SMR projects by tech companies near completion, aiming to be operational by 2030. Regulatory hurdles persist, but private lobbying expedites approvals in tech-friendly regions (e.g., the U.S., Canada, and parts of Europe).
   * Early SMRs are integrated with tech data centers, powering them sustainably and proving the feasibility of nuclear energy in a distributed grid.
* **AI for Renewable Energy Scaling**:
   * AI robots are deployed to construct and maintain wind turbines, solar farms, and geothermal facilities with high efficiency, even in remote areas.
   * AI-managed energy marketplaces emerge, allowing regions with surplus energy (from nuclear or renewables) to sell directly to energy-scarce areas.
* **Circular Economy Manufacturing**:
   * AI and robotics significantly reduce waste in manufacturing. Autonomous systems disassemble and repurpose old electronics, textiles, and building materials for reuse in new products.

2029-2032: Mainstream Adoption & Results

* **Commercial SMRs Operational**:
   * The first wave of Big Tech-funded SMRs goes online, producing cost-competitive, zero-emission energy. Major tech companies are largely energy self-sufficient and sell surplus nuclear energy to national grids.
   * AI systems manage nuclear operations, increasing safety and efficiency while reducing human oversight needs.
* **AI Robots Tackle Carbon Capture & Recycling**:
   * AI-powered robotics scale up direct-air carbon capture, making the process faster and more cost-effective.
   * Autonomous systems are deployed globally to clean up plastic waste and pollutants from oceans, landfills, and urban environments.
* **Manufacturing Transformation**:
   * Fully automated factories powered by nuclear energy and AI robots become standard for Big Tech and forward-looking nations. Emphasis is placed on reducing global emissions by producing only as much as needed and recycling nearly all waste.

2033-2040: Global Impact & Next-Gen Innovations

* **Decentralized Nuclear Grids**:
   * Big Tech's SMRs and advanced AI energy distribution systems become a global standard, especially in developing nations where energy demand is high, and governments struggle to fund renewable projects.
   * Intergovernmental bodies begin adopting Big Tech's nuclear models after years of bureaucratic delays with renewable-only strategies.
* **AI-Driven Climate Mitigation**:
   * AI robots and drones are used for large-scale reforestation, soil regeneration, and ocean algae farming to sequester carbon and restore ecosystems.
   * Climate-adaptive infrastructure, designed and built autonomously by AI, helps cities withstand extreme weather events.
* **Manufacturing as a Circular Economy**:
   * AI-led global manufacturing reaches near-zero waste as robots fully recycle old materials into new products without external energy inputs beyond clean nuclear.

Contrast with Intergovernmental Efforts:

* **COP Progress**:
   * While COP29 (2024) and subsequent summits lead to agreements, implementation delays persist due to competing national interests, limited funding, and slow adaptation of climate goals.
   * By 2030, COP-led initiatives like renewable energy targets and emission reductions fall short of expectations due to political resistance, uneven commitments, and bureaucratic inertia.
* **Big Tech vs. COP**:
   * Big Tech advances in SMRs, AI, and robotics outpace COP's progress in emission reductions. By 2040, Big Tech becomes the primary driver of global decarbonization, with governments relying on private-sector innovations to meet climate goals.

This timeline illustrates how **Big Tech could dominate climate solutions** while governmental efforts remain constrained by politics and bureaucracy. However, collaboration between the two—leveraging Big Tech's speed and innovation with government policies for equity and regulation—could create a more balanced and effective approach to addressing climate change.

Comments

u/HotHamBoy • 3 points • 2024-12-16 16:06 UTC

The most plausible scenario is that nothing meaningful happens and all the rich people move to the most hospitable places and create manufactured “paradises” while the rest of the world devolves into hell.

Humanity will survive as a smaller, more miserable population. Wars will be waged over resources. Many people will suffer and die, but we'll keep trucking and in some places humanity will be more or less living “normal,” just very few, wealthy places.

That's the most plausible scenario

> **u/Lurkerbot47** • 1 points • 2024-12-16 19:42 UTC
>
> We are not going to tech our way out of this, that's for sure. Best that can do is soften the landing.

u/tomttttttttttt • 3 points • 2024-12-16 15:05 UTC

If solar/wind continues to fall in price, and both iron-air and solid state batteries come through in the very near future as they both threaten to, then our path to decarbonisation becomes a whole lot easier and cheaper than it looks right now. Market forces will combine with increasing needs and desires to curb climate change to bring about the change we need.

u/hi65435 • 1 points • 2024-12-16 16:14 UTC

I mean for AI training not even a battery is needed. Different story for interference but that also consumes much less energy

u/tomttttttttttt • 2 points • 2024-12-16 16:18 UTC

batteries are needed to cover the intermittency issue of renewables, to allow us to reliably use that cheap and abundant energy source.

I'm not sure about your comment, I guess you can do AI training only when there's excess electricity being produced, I don't know what "interference" is in the context of AI? Do you mean when people are using LLMs and the like?

u/hi65435 • 1 points • 2024-12-16 16:54 UTC

>batteries are needed to cover the intermittency issue of renewables, to allow us to reliably use that cheap and abundant energy source.

Yeah but every modern power grid already has plenty of redundancies. From conventional power stations which can be regulated continuously and rarely run at 100%, gas power stations to Diesel generators which can be fired up almost instantly. This problem is as old as power grids are. FWIW Smart Grids take this whole approach to the extreme.

In fact such redundancies can be even there at the DC level and even server level (UPS)

>I'm not sure about your comment, I guess you can do AI training only when there's excess electricity being produced, I don't know what "interference" is in the context of AI? Do you mean when people are using LLMs and the like?

Exactly, training is 99% of power usage and interference is querying the trained model

u/tomttttttttttt • 1 points • 2024-12-16 19:11 UTC

Can't keep using gas, oil or coal.

u/hi65435 • 1 points • 2024-12-17 10:17 UTC

My main point being that it makes sense to use what's there and not wait another 5-10 years. Batteries are not really present on a grid level in most countries. On the other hand solar and wind are installed on a continuous base and are cheap.

It sounded the concern are black outs or brown outs. But I think this is very unlikely, even without having to use backup capacity when properly timing resource usage.

To be honest, to me this seems like a prime example of tech bro hubris. Planning to build and operate Nuclear reactors at a time when even Energy operators clearly state that they aren't economically viable anymore. When in fact there's already a solution which is mostly software based.

u/tomttttttttttt • 1 points • 2024-12-17 13:10 UTC

This is the r/futurology sub, if we can't talk at least 5-10 years into the future, what is the point of this sub?

It's not like the tech I'm talking about is sci-fi either - iron-air batteries go online next year, Toyota and Nissan are talking about 2028 for EVs with solid state batteries. The latter will probably slip but it's close enough now that it's almost certain to happen in the near future. It's iron-air batteries that are the game changer for grid storage anyway. If these work, it'll make thermal plants economically unviable.

I don't know why you think black and brown outs are unlikely on grids moving to 100% renewables generation - this is a massive concern for every grid operator around the world as far as I know - it definitely is in the UK. Just how do you expect to achieve this without backup capacity and without fossil fuels?

Because remember the topic of this particular post was about climate change and how to stop it, which means stopping using fossil fuels (unless you want to rely on an unproven tech in direct air CCS)

I'm not talking about nuclear so I don't know where you've pulled that out from. I'm talking about solar/wind plus battery storage.

I find it laughable that you call out tech bro hubris and then in the next sentence claim there's a software solution already in existence like software isn't tech?

Please do expand on this, I'd love to hear how a software solution can cover a couple of weeks of low wind in the north sea to provide electricty for northern Europe mid winter.

u/hi65435 • 1 points • 2024-12-17 13:36 UTC

My point was mainly about the intersection of the AI and Nuclear Aspect, considering that a high ratio of the energy use is batch jobs which *can be timed*

Not even Nation states running labs literally full of rocket scientists cannot guarantee the safety of NPPs, how should we be confident that Google or Microsoft do this? And yes, they are *planning* this today to be deployed in 2030. Indeed this is reasoning about the future and quite some hubris...

Why should we care? Because at least as a consumer or smaller business I'm able to run queries against AI models on whatever power source I like to. However it's complacent since the batch jobs are running against various power sources and probably eventually mainly Nuclear

u/quadrupedalism • 1 points • 2024-12-16 17:50 UTC

Inference not interference

u/OkToe7809 • 1 points • 2024-12-16 15:34 UTC

Those are all valuable energy sources to continue developing. The thing is, energy demand will grow exponentially with consumer demand for (gen) AI tech. The only energy source with any potential to meet that is nuclear, hence the need to resume its R&D - keep me honest.

u/tomtttttttttttt • 1 points • 2024-12-16 15:57 UTC

The sun is the biggest energy source we can imagine.

Current nuclear tech is not renewable in that uranium will run out, with exponentially growing demand, how long do you think we'll actually have before we run out of fuel? The sun will burn to the end of the solar system.

90 years on conventional reactors apparently: https://world-nuclear.org/information-library/nuclear-fuel-cycle/uranium-resources/supply-of-uranium

"The world's present measured resources of uranium (6.1 Mt) in the cost category less than three times present spot prices and used only in conventional reactors, are enough to last for about 90 years."

and that's the world nuclear association, and talking fuel up to three times the current cost - I don't know how that would translate into higher electricity costs because I don't know what portion of the cost of nuclear is fuel rather than other costs but as the cost of solar/wind falls and nuclear rises, then the viability of nuclear as a solution becomes lower and lower.

You can add in breeder reactors to this but they are essentially theoretical at this point so we don't know how much they will be able to reduce usage/re-use fuel. In theory it could be a factor of 100 so that 90 years becomes 9,000 years which is pretty good, but not *until the end of the solar system* good.

But how much of that 100x factor is going to be used up by the exponential demand growth you foresee? will we be back to a few centuries?

Perhaps we'll have mastered nuclear fusion in that time...

Still, if we're going to start talking about fusion we can also talk about orbital solar arrays and even the sun surrounding power generation thing whose name I am completely forgetting right now.

Besides which if we're talking practical responses, nuclear is too expensive and takes too long to build. If we get good cheap reliable storage then wind/solar is the way to do it because market forces are the easiest way to create mass change.

That said I do think that we should continue to fund R&D into SMRs, breeder reactors and especially fusion but in the next 25-50 years, it's solar/wind and storage that is the best path to decarbonisation **IF** iron-air and solid state batteries prove their mettle, which is likely in the next few years (next year for the first grid scale iron-air battery, 2028 for solid state battery EVs from Nissan and Toyota, expect the latter to slip but the iron-air battery will come online next year and either work or not...) and nuclear can be dropped entirely (and will be due to the cost and time requirements).

u/michael-65536 • 1 points • 2024-12-17 03:01 UTC

Uranium running out is something of a red herring. This kind of timescale is entirely normal, relatively long even. Without even considering other empirically proven fuel cycles, or reprocessing nuclear fuel rather than wasting the vast majority of it like we usually do, the estimated proven reserves are largely a product of how much we've bothered looking for, which is a product of demand.

To quote the website you linked; "This represents a **higher** level of assured resources than is **normal** for most minerals."

Nuclear has plenty of legitimate drawbacks, that's not really one of them.

u/tomtttttttttttt • 1 points • 2024-12-17 06:03 UTC

Except that the new reserves we move to get progressively more expensive making nuclear less and less viable as we run through the uranium compared to nice and cheap solar and wind.

Breeder reactors are unproven afaik. Reprocessing fuel extensively is not a given tech right now is it?

u/michael-65536 • 1 points • 2024-12-17 14:21 UTC

To the extent that it happens with every raw material used in every manufactured product which exists, yes that will happen with some of the new reserves. Some will alos turn out to be cheaper. This will also happen with everything you need to make pv cells and wind generators.

Breeder reactors and reprocessing have already been invented and used decades ago. They're not widespread at the moment because raw uranium ore is so abundant (and also the mining lobby has significant political lobbying influence).